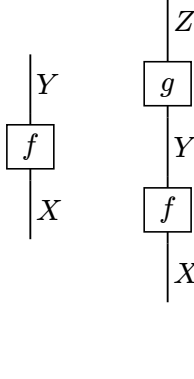
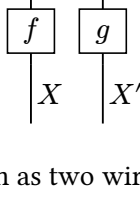


copycat gallery

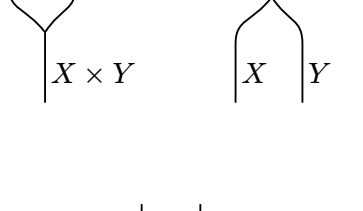
A kernel $f : X \rightsquigarrow Y$, and sequential composition:



Parallel composition:



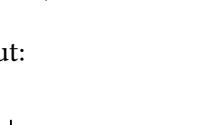
A product wire $X \times Y$ may be drawn as two wires, and back:



The swap:



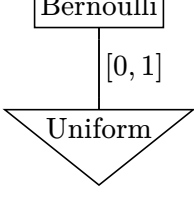
The Dirac kernel is the bare wire:



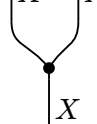
A state is a kernel with trivial input:



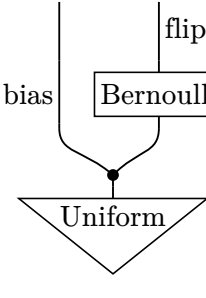
A little probabilistic program:



Copying:



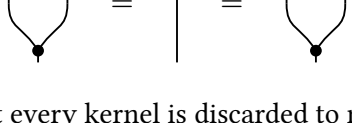
Sampling a bias, then flipping a coin with it:



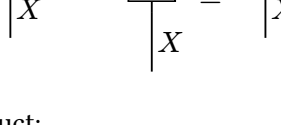
Coassociativity and cocommutativity of copying:



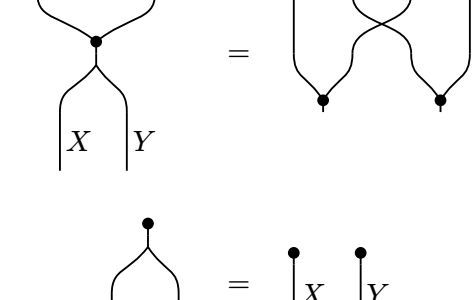
Counitality:



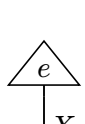
Discarding, and the fact that every kernel is discarded to nothing:



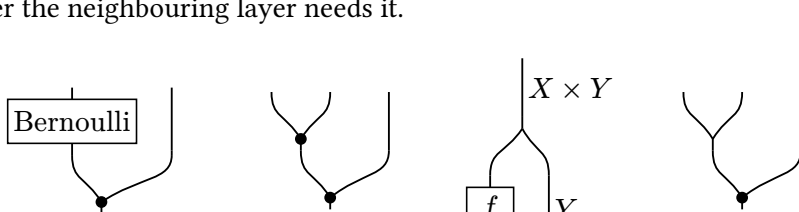
Copying and discarding a product:



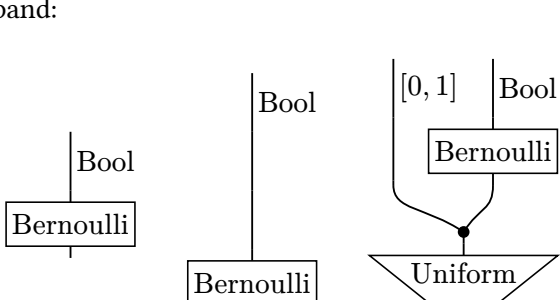
An effect (the mirror image of a state):



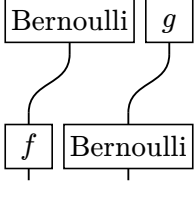
The arms of copy, unbundle and bundle are flexible: each runs straight from its dot to wherever the neighbouring layer needs it.



Plain wires are flexible too. They follow the box they stand on, so none of these needs a connector band:



A wire whose two ends are held at different positions bends by itself, with no band inserted:

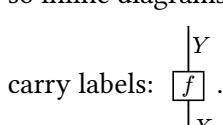


Inline diagrams

The copy map $\vee$ and the discard map $\bullet$ satisfy $\vee \bullet = \text{identity}$, so every object is

a comonoid. At the default unit the same term, $\vee \bullet$, is too tall for a line of text,

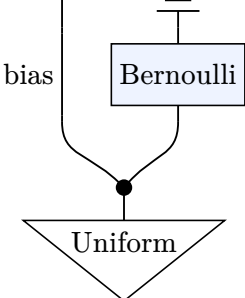
so inline diagrams should pass a smaller unit, and a smaller label size when they



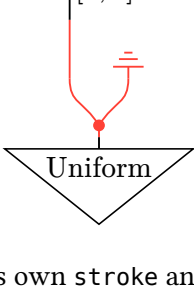
carry labels: $\boxed{f} : X \rightarrow X$.

Style overrides

A style can be set for the whole diagram, through the style: argument of string-diagram. Element groups inherit the root stroke and fill, and partial strokes fold as in cetz:



It can be set for a sub-diagram, through styled or the style: argument of serial and parallel:

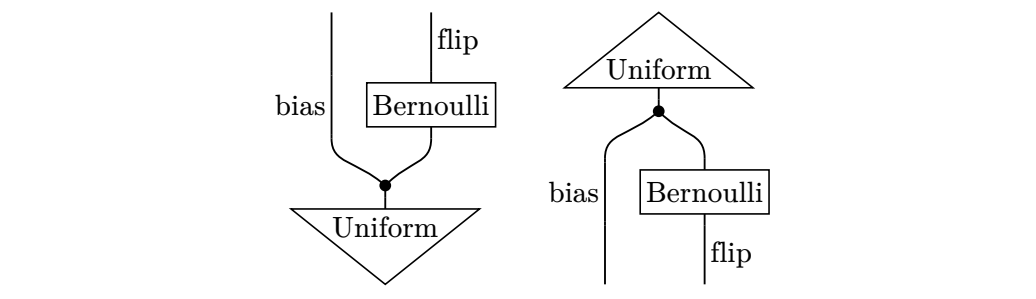


Or for a single element, through its own stroke and fill arguments:



Reading direction

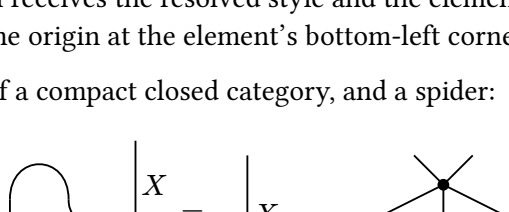
The style key direction reads the same diagram bottom to top (the default), top to bottom, left to right, or right to left. Read sideways, a wire label with side: "right" sits below its wire and one with side: "left" above it.



Custom primitives

primitive turns a cetz drawing into a rigid element that composes like any other. The drawing function receives the resolved style and the element's geometry, and draws in units with the origin at the element's bottom-left corner.

The snake equation of a compact closed category, and a spider:



An element can size itself to its label by giving width or height as a function of the style and a measure function. measure reports the label's extent along the element's own width and height, so this trapezoid fits its label in every reading direction:

